
From Data to Decisions: Machine Learning in Finance and Business

Bachelor's diploma seminar: introduction and rules

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Agenda

1. The seminar
2. How we work
3. Timeline and passing
4. Faculty rules in brief
5. Think bigger: articles and teams
6. AI tools and academic integrity
7. Topics
8. Writing standards
9. Next steps

What this seminar is for

Course 2400-PL3SL339A • 3 ECTS per semester • pass/fail

You write a bachelor's thesis that uses **modern quantitative methods** (machine learning, statistical learning, econometrics) to answer a well-posed question in **finance or business**.

What I care about most:

- a research question that matters to someone, stated precisely,
- rigorous validation: honest baselines, out-of-sample testing, robust conclusions,
- results that anyone can reproduce from your repository,
- a coherent story: *question* → *data* → *model* → *decision or conclusion*.

Applying a method is the easy part. Interpreting and criticising the result is the thesis.

Thematic scope

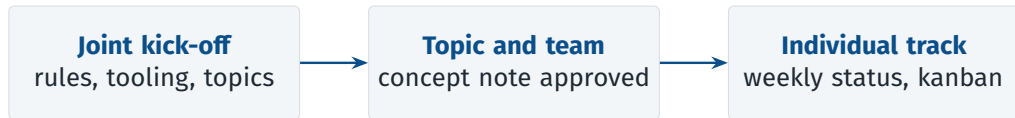
Deliberately broad: any topic with a sound question and a rigorous quantitative method

- Comparing quantitative methods on a financial or business problem
- Case studies: finance, insurance, e-commerce, logistics
- Probabilistic modelling and forecasting
- Risk measurement and uncertainty
- Causal inference for business decisions and policy evaluation
- ML beyond classic analytics: behavioural economics, public policy

Where I can help most

The intersection of econometrics and machine learning: quantitative finance, risk modelling, time-series forecasting, probabilistic methods, customer analytics. I combine academic research with commercial projects in insurance, e-commerce and logistics.

Format: remote, one joint start, then your own track



- All meetings are **remote, on Google Meet**. This is the preferred way to reach me.
- After the kick-off we work individually or in project teams, not as a lecture group.
- Joint sessions return at two points: *research design* and *interpreting results*, where you present and get feedback from the group.
- Asynchronous questions go to **GitHub issues** in your project repository, so the answer stays next to the work. University e-mail is for formal matters.

Weekly status and a kanban board

Every project lives in a GitHub Project. The board is the single source of truth for progress.



- **Every week: a status check-in.** What moved, what is blocked, what is next.
- I read the board before we talk, so the call is about decisions, not reporting.

Kanban: the working rules

Cards

- One card = one GitHub issue = one piece of work you can finish in days, not weeks.
- Each card states its **definition of done**: a figure, a table, a section, a passing test.
- Writing is work too: chapters and sections are cards.

Flow

- **Limit work in progress**: at most two cards *In progress* per person.
- Work happens on branches; a pull request linked to the issue moves the card to *In review*.
- I review there. Merged means *Done*.
- **No movement on the board for weeks is the main risk signal** for passing the semester.

The toolchain is part of the course

These are learning outcomes in the syllabus, not preferences

Writing **LaTeX only** (Overleaf or a local TeX distribution); bibliography in BibTeX.

Versioning Everything in **git**, hosted on GitHub: thesis sources and analysis code, with a readable commit history.

Language **Python** preferred; R accepted.

Environment Python: `uv` with `pyproject.toml`. R: `renv`.

Standard **Reproducibility**: anyone with access to the repository can recreate your results without contacting you.

New to some of these? That is expected. The first weeks include time to set them up, and the seminar repository has a thesis template and links to get started.

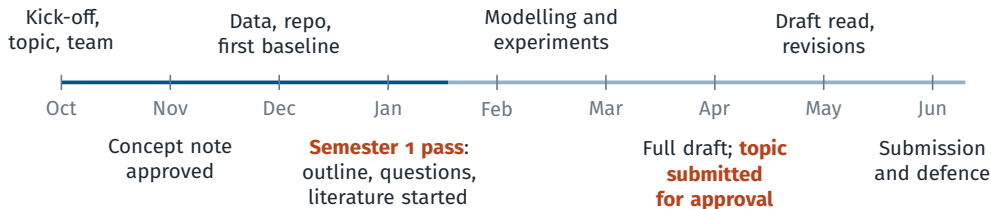
What your project repository should look like

my-thesis/	
README.md	how to reproduce
pyproject.toml	uv environment
uv.lock	
data/	raw data, fetch script
src/	reusable code
experiments/	configs, run scripts
results/	generated output
paper/	LaTeX sources, .bib

- One command rebuilds the results; one command rebuilds the PDF.
- Figures and tables are **generated by code**, never pasted by hand.
- Fix random seeds; record data versions and download dates.
- Data you cannot share: document how to obtain it and commit a small synthetic sample.
- No secrets, API keys or personal data in git, ever.

Indicative timeline

Exact submission and defence dates follow the WNE academic calendar



- Give me **time to read**: a full draft needs weeks, not days, before any deadline.
- The schedule is tight only if modelling starts in spring. Start in December.

What it takes to pass each semester

After semester 1

- thesis outline in LaTeX,
- data collected,
- research questions and hypotheses,
- literature review started,
- git repository with thesis sources and analysis code.

After semester 2

- literature review and modelling complete,
- thesis ready for defence,
- code reproduces all results from the repository,
- for high-potential work: a draft of a scientific article.

In both semesters: **documented progress** (commits, successive versions of the document) and **active participation** in consultations and joint discussions.

Source: course syllabus, USOS 2400-PL3SL339A

Your topic must be formally approved

Procedure of the WNE Teaching Council (Rada Dydaktyczna)

1. We agree the topic together. I am responsible for checking that it fits your programme, level of study and leading discipline.
2. I submit it to the Dean's Office during the last semester of the seminar, **no later than 4 weeks before you plan to submit the thesis.**
3. The Teaching Council reviews the topic within 14 days. It may request changes, after which the corrected topic is submitted again.
4. **A thesis without an approved topic cannot be entered into APD or defended.**

In practice: we fix the final title early in spring, and we word it so the economic or financial problem is visible in the title itself.

Source: Komunikat nr 1/2025 of the Chair of the WNE Teaching Council, 20 May 2025

What WNE expects from a bachelor's thesis

- A **clearly defined research goal**; it may include a research hypothesis.
- Evidence that you can do research using current Polish and international literature.
- Research tools consistent with your programme, or going beyond it.
- Either your own study, or an in-depth literature-based report. In this seminar we do **empirical work**.
- Traditional form: up to about **50 pages** (100,000 characters with spaces). Concise beats long.
- On a Polish-language programme the thesis may be written in **Polish or English**.
- It may take the **traditional form or the form of a scientific article**.

Source: wne.uw.edu.pl → Student → Prace dyplomowe → Wytoczne dotyczące prac

The method must serve an economic or financial problem

How the Teaching Council judges whether a topic fits the programme

Programmes	The thesis should
Economics (Committee 1)	analyse economic phenomena at micro or macro scale, build on economic theory or policy, and pose a research problem grounded in the literature.
Finance, accounting, quantitative finance (Committee 2)	address a genuine finance question (pricing, risk management, portfolio optimisation, ...). Statistical, econometric and ML methods are welcome <i>as long as they serve the financial problem</i> ; a generic “business” theme is not enough.
Informatics and Econometrics, data science (Committee 3)	show independent, critical use of advanced quantitative methods, state the economic, social or financial context, and interpret results in those terms.

For us: “I benchmarked five models” is not a topic. “Which models forecast portfolio tail risk reliably enough for capital decisions” is.

Source: Komunikat nr 1/2026 of the Chair of the WNE Teaching Council, 23 March 2026

Formal requirements: the template handles most of them

Required order

1. Title page
2. Signed declarations: supervisor and author(s)
3. Abstract (max. 800 characters), keywords, Erasmus code, English title
4. Table of contents (max. three levels)
5. Main text
6. Bibliography
7. Lists: abbreviations, tables, figures, appendices

Typesetting

- Times-like font, 12 pt; line spacing 1.5
- All margins 25 mm; page numbers bottom centre
- Tables and figures numbered through the whole thesis, each with a title and a **source**
- Citations as (Author, year); bibliography alphabetical, preferably **APA**

Source: Załącznik B and Załącznik D to Resolution no. 7 of the WNE Teaching Council

From finished thesis to defence

1. **Submit** the thesis file by e-mail to the Dean's Office; they open your entry in APD.
2. **Upload** a searchable PDF to APD with the English title, abstract and keywords, then send it for the supervisor's approval.
3. **Clear your accounts:** IT lab, WNE library, BUW, language and PE units.
4. **Anti-plagiarism check** in JSA, run by the supervisor; I accept the thesis after reviewing the report.
5. A **reviewer** is appointed. Both reviews are uploaded to APD at least 3 days before the defence.
6. One **printed copy**, double-sided, soft-bound, with the AI declaration, delivered no later than 2 days before the exam.

For June and September defences the submission deadline is strict to the hour (12:00).

Source: wne.uw.edu.pl → Prace dyplomowe → Składanie prac; Załącznik C

How the thesis is graded

- 5! outstanding: well beyond the programme, a real contribution to science
 - 5 very good: makes significant use of the programme's content, fully meets requirements
 - 4 good: moderate use of the programme's content
 - 3 weak: sufficient use, requirements partly met
 - 2 fail: substantive errors leading to wrong conclusions
-

- Supervisor and reviewer each score the thesis on the official review form; half grades are allowed.
- **Final grade = arithmetic mean** of the two.
- The reviewer sees only the thesis. Everything that matters has to be on the page.

Source: wne.uw.edu.pl → Prace dyplomowe → Ocena prac dyplomowych

The thesis can be a scientific article

This is the route I encourage for ambitious projects

- The article has the structure typical for its field, about **10–12 thousand words**, formatted for a chosen journal.
- A **supplement** is mandatory: literature review, methodology and results in more depth. It guides the reviewer and documents your own contribution.
- Article and supplement may be in different languages.
- Publication before the defence is **not required**.
- The supervisor may co-author the article; the supplement is the student's own work.

Layout of an article-form thesis

Title page → declarations → abstract page → contents → “Scientific article” → article → “Supplement” → supplement → lists

Source: wne.uw.edu.pl → Prace w formie artykułów naukowych; Załącznik B, point 17

Co-authorship: what the rules allow

Form	Students	Each student	Supervisor
Traditional thesis	2	exactly 50%	–
Scientific article	1	at least 60%	at most 40%
Scientific article	2	at least 40%	at most 20%

- A thesis can have **at most two student authors**.
- Every author declares their substantive and percentage contribution, chapter by chapter.
- I must justify the co-authorship in my review.

Source: WNE thesis guidelines; declarations in Załącznik D

Larger projects, several theses, one paper

The model

- A research programme bigger than one thesis, split into **clearly separable parts**.
- Each part is one thesis, by one student or a pair, with its own question and contribution.
- Shared infrastructure: one repository or organisation, common data pipeline, common evaluation protocol.
- The parts combine into a **journal submission**.

Why bother

- A stronger benchmark than any one person can build.
- You learn to work the way research and industry teams do: issues, reviews, shared code.
- A paper and a public repository are worth more on a CV than a grade.

Authorship follows contribution, agreed and written down at the start of the project.

AI tools: allowed, declared, and your responsibility

Faculty requirement

Every thesis includes a signed **declaration on the use of AI systems**: which tool you used and why, or that you used none. You confirm that you verified the content and take full responsibility for the final thesis.

Working rules in this seminar

- Use AI like a capable colleague: code, debugging, language, critique. Not to produce the research idea, the interpretation or the conclusions.
- **You must be able to explain every line of code and every sentence.** I will ask.
- **Verify every reference.** Citing a paper that does not exist is a disqualifying error.
- Keep a short log of what you used AI for; the declaration then writes itself.

How to propose a topic

First message: three things

1. A working title.
2. Two or three sentences: whose problem is it, what do they want to achieve, what is missing today?
3. Confirmation that **the data exists** and you can get it.

Your own idea is welcome. So is joining one of the projects on the next slides.

Then: a two-page concept note

1. The problem
2. Why it matters to an economist or a financial decision-maker
3. Research questions and hypotheses
4. Methods and data
5. Planned structure

The concept note later grows into your introduction. It is the first card on your board.

Project 1: foundation models for market risk

Task-specific heads on Chronos for volatility and tail-risk forecasting

Idea. Chronos models are time-series foundation models pretrained on large, mostly non-financial corpora. Keep the backbone and train lightweight **distributional or quantile heads** on its representations to forecast volatility, Value-at-Risk and Expected Shortfall.

Questions

- Does generic pretraining transfer to heavy-tailed, volatility-clustered returns?
- Zero-shot, new head on a frozen backbone, or fine-tuning: what is worth the compute?
- Do forecasts pass regulatory-style backtests, not only win on average loss?

Splits naturally: volatility | VaR and ES | asset classes | fine-tuning strategies.

Benchmarks and evaluation

- GARCH family, HAR-RV, quantile regression
- VaR coverage and independence tests
- joint VaR-ES scoring, model confidence sets

Project 2: detrending and deseasonalisation

A review and benchmark: what to do about non-stationarity before modelling a time series

Idea. Every forecasting pipeline makes a choice about trend and seasonality, usually by habit. Build a **systematic review and a controlled benchmark** of these choices.

Methods to map

- differencing and transformations
- filters: Hodrick–Prescott, Hamilton, band-pass
- decompositions: STL, MSTL, X-13, TRAMO-SEATS
- Fourier terms, wavelets, EMD, VMD
- learned: RevIN, neural decomposition blocks

Questions

- When does preprocessing help, and when does it destroy signal or leak the future?
- Do ML, deep and foundation models still need it?
- How do conclusions change across macro, financial and retail demand series?

Splits naturally: review and taxonomy | benchmark on public datasets | economic case study.

Project 3: foundation models for tabular data

A review of pretrained models for tabular prediction, with economic applications

Idea. Gradient-boosted trees have ruled tabular data for a decade. Pretrained **tabular foundation models** (in-context learners such as the TabPFN line, LLM-based and cross-table approaches) now challenge them. Review the field and test the claims where economists need them.

Questions

- In which regimes do they win: small samples, many categories, distribution shift?
- Are the predicted probabilities **calibrated** well enough for pricing and credit decisions?
- What about interpretability, cost and regulatory constraints in finance and insurance?

Applications. Credit scoring, insurance pricing, churn and customer value.

Splits naturally: literature review | benchmark vs. boosting | calibration and interpretability.

The logic of a thesis



- Each step follows from the previous one. If a chapter does not help answer the question, it does not belong.
- **Introduction:** problem, why it matters, hypotheses, methods and data, structure.
- **Every chapter:** what it does and why; the content; what was established and how it leads to the next chapter.
- **Data chapter:** how the data was generated and collected, exact meaning of variables, descriptive statistics, and what this implies for modelling.
- **Conclusions:** your results, limitations, practical use, future work.
- **Abstract:** understandable to someone who will never read the thesis.

Style: a professional document, not literature

Text

- Short sentences, precise words. No metaphors, no “some”, “many”, “appropriate”.
- Claims need evidence: a citation or your own result. “It is well known” is neither.
- Cite what is relevant to *your* question, not everything the paper says.
- Draw only the conclusions your results support. State limitations plainly.

Figures and tables

- Every one is referenced by number and **interpreted in the text**.
- The caption says what the reader should see, and the source is always given.
- Readable at print size; consistent notation, units and model names throughout.
- No screenshots of code or output. Tables are typeset, plots are vector graphics.

Before our next meeting

1. Create a GitHub account (if you do not have one) and send me your username.
2. Read this deck and the docs/ folder in the seminar repository.
3. Install the toolchain: git, a TeX distribution or Overleaf, uv or renv.
4. Build the **thesis template** from the repository once, to check that your setup works.
5. Send your topic proposal: title, two or three sentences, data confirmation. Say whether you want to work alone, in a pair, or join a larger project.
6. Book your weekly status slot on Google Meet.

Everything is in one place

github.com/wne-uw-mjwozniak/wne-uw-seminars